

BAIT3153 SOFTWARE PROJECT MANAGEMENT

Question 1

ABC Tech Solutions, a mid-sized software development company specialising in custom software solutions for various industries, has been approached by HealthPlus, a healthcare provider, with 20 branches in Malaysia, to develop a comprehensive patient management system (PMS). The system aims to streamline patient data management, appointment scheduling, billing, and telemedicine services. HealthPlus provided several key requirements for the PMS, including secure storage of patient medical histories, an intuitive interface for booking and managing appointments, an automated billing system with insurance processing and online payments, and a secure telemedicine feature for virtual consultations. The system also needs to support different user roles to ensure data privacy and security, and comply with healthcare regulations such as HIPAA (Health Insurance Portability and Accountability Act). It is anticipated that there will be high risks with the system installation as it involves the entire nation.

The project has a duration of 12 months and a budget of RM500,000. The project team includes John Smith as the Project Manager, Alice Johnson as the Lead Developer, Sarah Lee as the UI/UX Designer, Mark Brown as the QA Specialist, and Emma White as the Business Analyst.

Answer the following questions based on the ABC Tech Solutions scenario.

- a) Identify **TWO (2)** major business goals and **ONE (1)** project goals. For each objective mentioned, elaborate on one effectiveness measurement that could be used to determine if the objective has been achieved. (12 marks)

Business Goals:

1. Ensure Compliance with Healthcare Regulations (e.g., HIPAA):
 - Effectiveness Measurement is to conduct regular compliance audits to ensure 100% adherence to HIPAA standards and no reported data breaches.
2. Improve Patient Service Efficiency:
 - Effectiveness Measurement is to measure reduction in average patient appointment scheduling and billing time by at least 40% after system implementation.

Project Goal:

- Deliver Patient Management System within 12 Months and RM500,000 Budget
- Effectiveness Measurement is to track project milestones against the project baseline (Gantt chart, Earned Value Analyze) to ensure completion on schedule and within allocated budget.

- b) Discuss **TWO (2)** importance of resource allocation and resource leveling in the context of the given project scenario. Identify **TWO (2)** challenges might ABC Tech Solutions face if these processes are not managed effectively? (4 + 2 marks)

Importance

1. Avoids Overloading Key Team Members:
 - Resource allocation makes sure work is shared fairly so no team member such as Lead Developer and QA Specialist are overburdened.
2. Improves Project Efficiency
 - Proper resource allocation ensures the right skills are assigned to the right tasks to improve workflow and prevent delays.

Challenges if Not Managed Effectively

1. Project Delays
 - Overloaded team members may not complete critical tasks on time which can cause project delay.
2. Quality Issues
 - Misallocation of resources may lead to unqualified resources handling complex tasks which can result in system defects.

- c) During the project execution, John Smith, the project manager, identified the issues related to task slippage and poor-quality deliverables. Identify **ONE (1)** corrective action for each identified issue and evaluate the potential drawbacks of each corrective action. (7 marks)

[Total: 25 marks]

1. Task Slippage
 - Corrective Action is Re-planning the project such as redistributing workload. This can bring the project back on track, aligns tasks with available resources, and reduces the risk of further delays.
 - For drawbacks, it may cause changes to the overall timeline, affect dependent tasks, and require extra effort to update stakeholders.
2. Poor-Quality Deliverables
 - Corrective Action is Increasing the number of inspections such as more frequent QA testing. Detects errors earlier, improves final product quality, and builds client confidence.
 - For drawbacks, it may cause increased time and cost for testing, may slow down development, and could frustrate developers if inspections feel excessive.

BAIT3153 SOFTWARE PROJECT MANAGEMENT**Question 2**

Answer the following questions based on the ABC Tech Solutions scenario from **Question 1**.

- a) During the requirement analysis, HealthPlus CEO, Mr. Alan, told Mr. John Smith, the project manager, “We hope to use certain parts of the system earlier than the full implementation to address our immediate needs. This will allow us to start managing patient data more efficiently while the remaining functionalities are still being developed.”

Identify a suitable software process model to be adopted for the above scenario. Justify **TWO (2)** reasons of your choice. (1 + 4 marks)

Incremental Model.

1. Early delivery of useful system parts
 - HealthPlus wants to use certain modules such as patient data management before the full patient management system is ready. The incremental model supports early deployment of working modules.
2. Flexibility to add remaining features
 - New modules such as billing and telemedicine can be developed in later increments, reducing project risk and allowing continuous feedback from HealthPlus.

- b) Establishing a proper quantitative standard to measure the software is crucial for ensuring its quality. By defining specific metrics, the system can be effectively assessed and improved.

- (i) Considering McCall's software quality attributes, explain **TWO (2)** possible *quality attributes* relevant to the HealthPlus Patient Management System. For each identified quality attribute, provide **TWO (2)** metrics that can be used to measure them.

(6 + 8 marks)

Reliability – The system must consistently perform critical tasks such as patient data storage and telemedicine without failures.

Metric 1: Mean Time Between Failures (MTBF)

– This measures the average operating time between one failure and the next. For example, if the PMS can run for 500 hours before experiencing a crash, then the MTBF is 500 hours. A higher MTBF means greater system stability and better user confidence in using the system daily.

Metric 2: Failure Rate

– This calculates the frequency of errors or crashes, such as the number of failed billing transactions per 1,000 transactions. It helps determine whether the system can handle a large number of patient records and transactions without frequent breakdowns. A lower failure rate indicates higher reliability and smoother patient management.

Security – Essential due to HIPAA compliance and patient privacy.

Metric 1: Number of Security Incidents Detected

- This tracks the count of unauthorized login attempts, malware attacks, or breaches reported within a given time. For example, fewer than 2 incidents per month would indicate strong security. A high number of incidents signals vulnerabilities that must be addressed immediately.

Metric 2: Percentage of Modules Passing Security Tests

- This measures how many system components such as patient data storage and online payments pass penetration testing or vulnerability scans. For instance, if 18 out of 20 modules pass, that's 90% compliance. A higher percentage shows that the PMS meets security standards and reduces risks of data leakage.

(ii) Discuss **TWO (2)** characteristics of good metrics.

(6 marks)

[Total: 25 marks]

1. Programming Language Independent

- Metrics should not depend on a specific programming language so that results remain comparable across projects and technologies.
- For instance, using Function Points (FP) or number of defects per module is independent of whether the system is developed in Java, Python, or .NET, ensuring fair measurement.

2. Simple and Computable

- A good metric should be easy to understand and calculate without requiring too much time or effort.
- For example, measuring failure rate per 1,000 transactions in the system can be done directly from system logs, making it practical for project monitoring.

Question 3

Answer the following questions based on the ABC Tech Solutions scenario from **Question 1**.

- a) During project execution, John Smith faced some difficulties in tracing the project progress. Justify **TWO (2)** reasons why software project's progress is not visible as compared to other types of projects. (6 marks)

1. Intangible Deliverables

- Unlike construction projects where progress is visible such as building walls, software progress is less obvious since coding, design, and testing are not physically observable.

2. Complex Dependencies

- Progress may not be clear because many tasks such as coding, integration, and testing rely on each other. If one part, such as the billing module, is delayed, the whole project may look stuck even though work is still going on.

b) John Smith would like to improve the software development process. He needs to carry out process measurement and process analysis before any process changes can be made.

(i) Elaborate **THREE (3)** process metrics which can be collected during process measurement. (6 marks)

1. Time Taken for Process Activities
 - Measures how long it takes to finish key tasks.
 - Example: Time to complete requirements gathering or UI design for the system.
2. Resources Required for Activities
 - Tracks the total effort and resources used.
 - Example: Hardware/Software used during testing.
3. Number of Occurrences of a Particular Event
 - Records how often certain events happen during the process.
 - Example: Number of run-time errors found during the first User Acceptance Test (UAT).

(ii) Based on the challenge faced by John Smith mentioned in **Question 3 a)**, suggest a goal for the Software Process Improvement (SPI). Elaborate on **TWO (2)** most significant process attributes to be considered during the SPI to achieve the suggested goal. (1 + 6 marks)

Goal: Improve software quality and ensure timely delivery of the Patient Management System.

1. Reliability
 - The process should be able to detect or prevent errors early, such as catching issues in patient data storage before they reach production. This improves compliance with healthcare standards (e.g., HIPAA) and reduces risks of system failure.
2. Rapidity
 - The process should support fast delivery of working modules, such as releasing the patient management module early while other parts are still in development. This helps HealthPlus use the system sooner and ensures progress is visible to stakeholders.

- c) System release is a crucial phase in the software development lifecycle where the developed software is made available to users. Proper release planning is essential to ensure the software is delivered on time, within budget, and with the desired quality. Discuss **TWO (2)** major factors that influence system release planning. (6 marks)

[Total: 25 marks]

1. Business & User Needs
 - HealthPlus may urgently need the patient data management module before others. Release planning should prioritize delivering this early.
2. System Quality & Stability
 - Only stable, well-tested modules should be released. For example, releasing billing or telemedicine features with security flaws could lead to compliance issues.

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Question 4

Answer the following questions based on the ABC Tech Solutions scenario from **Question 1**.

- a) One of the identified risks during the risk analysis activity is project slippage. Propose a suitable plan for risk mitigation, monitoring, and management actions to resolve this issue. (9 marks)

1. Risk Mitigation
 - Break the project into smaller milestones with clear deadlines.
 - Assign additional resources to critical tasks to reduce bottlenecks.
2. Risk Monitoring
 - Conduct weekly progress reviews with the project manager and team leads.
 - Use project management tools (e.g., Gantt charts, Jira) to monitor delays.
3. Risk Management Actions (if slippage occurs):
 - Reallocate staff to high-priority modules.
 - Negotiate with HealthPlus to adjust scope or delivery sequence such as release patient data management module first.

- b) (i) Dependability is one of the important properties for a system. Based on the given scenario, evaluate if the HealthPlus Patient Management System is a *safety-critical* or *business-critical* system. Provide **TWO (2)** justifications for your answer.
(1 + 8 marks)

Business-critical system.

1. Impact on Operations not Direct Patient Safety
 - The PMS manages appointments, billing, and telemedicine. A failure affects business efficiency and service delivery but does not directly put patients' lives at risk, unlike life-support machines.
2. Financial and Regulatory Consequences
 - Failure may cause billing errors, data loss, or HIPAA violations, which could result in legal penalties and financial loss, but it does not immediately endanger patient safety.

- (ii) Propose **ONE (1)** suitable approach to achieve dependability for the HealthPlus Patient Management System. Provide **ONE (1)** justification for the selected approach with **ONE (1)** appropriate example.
(7 marks)

[Total: 25 marks]

Implement Redundancy and Backup Systems such as failover servers. This ensures patient records, appointments, and billing information remain available even if one server or module fails to maintain system reliability and user trust. For example, if the main database server crashes, an automatic switchover to a backup server allows HealthPlus to continue accessing patient records without downtime.